

QUAID-I-AZAM UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE (BS 5th)

Terminal Examination Fall, 2020

Course No. ST-101 Course Title: Probability and Statistics

Max. Marks: 50

Time Allowed: 3 Hours

Teacher's Name: Mr. Abdullah Iqbal

Date: 15-JAN-2021

Note: Please attempt five questions. All questions carry equal marks.

- Q.1 (a)** Write down the difference between the operations of Poisson and Normal distribution? **(02)**
- (b)** A T.V. dealer extends credit to people buying his sets. Assuming in past 10% of all those he extended credit did not pay and he took a loss on each such set and made profit on remaining sets. Assume further that he sells 10 sets to 10 individuals independently on credit. If, buyer pays in full he makes a profit of Rs. 100. If, buyer does not pay he takes a loss of Rs. 200. Find expected profit of this dealer on these 10 sales. **(04)**
- (c)** If on the average rain falls on twelve days in every thirty, find the probability that **(i)** the first three days of a given week will be fine and the remaining wet, **(ii)** rain will fall on just three days of a given week. **(04)**
- Q.2 (a)** How would you define Measure of Central Tendencies? **(02)**
- (b)** The mean age of an individual boy is denoted by b , and the exact age of an individual girl is denoted by g . Exactly 5 years ago, the sum of the ages of 10 boys was 127.0 years, so $\Sigma(b-5) = 127.0$. In exactly 5 years' time, the sum of the ages of 15 girls will be 351.0 years, so $\Sigma(g+5) = 351.0$. Find the mean age today of **(04)**

- (i)** The 10 boys
(ii) The 15 girls
(iii) The 10 boys and 15 girls

(c) Find A.M., Mode, Q_1 and Q_3 from the given data. **(04)**

Marks	Above 95	Above 90	Above 85	Above 80	Above 75	Above 70	Above 65	Above 60	Above 55	Above 50	Above 45	Above 40
No. of Students	195	417	697	952	1231	1521	1791	1991	2152	2299	2432	2500

- Q.3 (a)** How could you define Poisson distribution as the limiting form of Binomial distribution? **(02)**
- (b)** Before buying a batch of 100 fuses, a man tests 4 of these. If the batch contains 10 defective fuses, what is the probability that he will find **(i)** no defective fuse in the sample? **(ii)** 4 defectives in a sample? **(04)**
- (c)** Given that the height of college boys has an average of 5'-2" and standard deviation of 4", and that the minimum height required for joining the N.C.C. is 5'-4", find the percentage of boys who would be rejected on account of their height? **(04)**
- Q.4 (a)** Describe a Poisson Distribution with the help of at least two real life examples? **(02)**
- (b)** Suppose that airplane engines operate independently and fail with probability of $1/5$. If a plane can make a successful flight with half of its engines working determine whether a 4-engine plane or a 2-engine plane has a chance of making a successful flight. **(04)**
- (c)** A manufacturer of cotter pins knows that 5 percent of his product is defective. If he sells cotter pins in boxes of 100, and guarantees that not more than 4 pins will be defective, what is the approximate probability that a box will fail to meet the guaranteed quality? **(04)**
- Q.5 (a)** The arithmetic mean and standard deviation of the series of 20 items were calculated by a student as 20 cm and 5cm. respectively. But while calculating them an item 13 was misread as 30. Find the correct values of the arithmetic mean and standard deviation? **(04)**
- (b)** It is known that 20 girls each have at least one brother. The number of brothers that they have is denoted by "x". Information about the values of "x-1" is given in the following table:

x-1	0	1	2	3	4
No. of girls	2	4	8	5	1

Use coded values to calculate standard deviation of number of brothers, to 3 decimal places. **(03)**

(c) The time (in minutes) that a random sample of 12 employees took to complete a task is:

13, 49, 40, 28, 23, 15, 17, 21, 37, 66, 30, 45

Compute the mean and the standard deviation for this sample. Also compute the percentage of observations lying within the limits $\bar{X} \pm S$. **(03)**

BEST OF LUCK